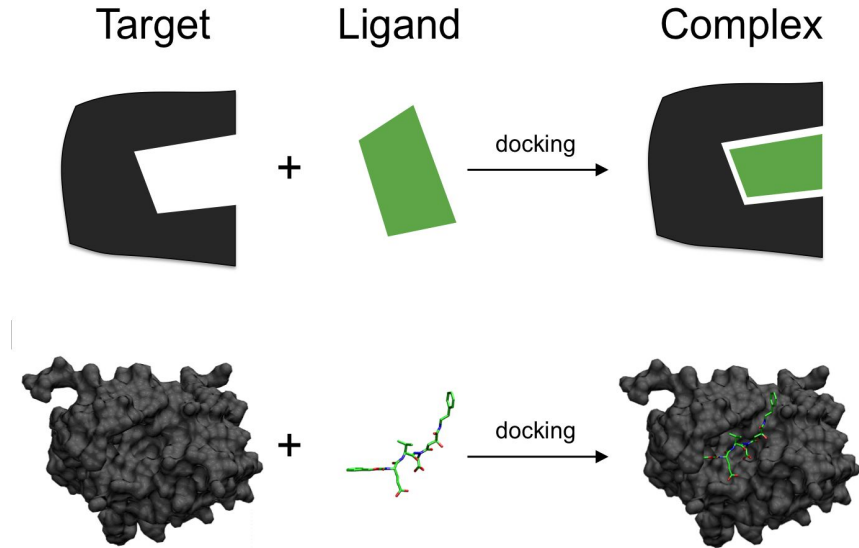


Molecular Docking Tutorial

Presented By:Diksha Marwaha

Course Instructor:Dr.Arjun Ray

Course -Biophysics(Bio215)



What is Molecular Docking?

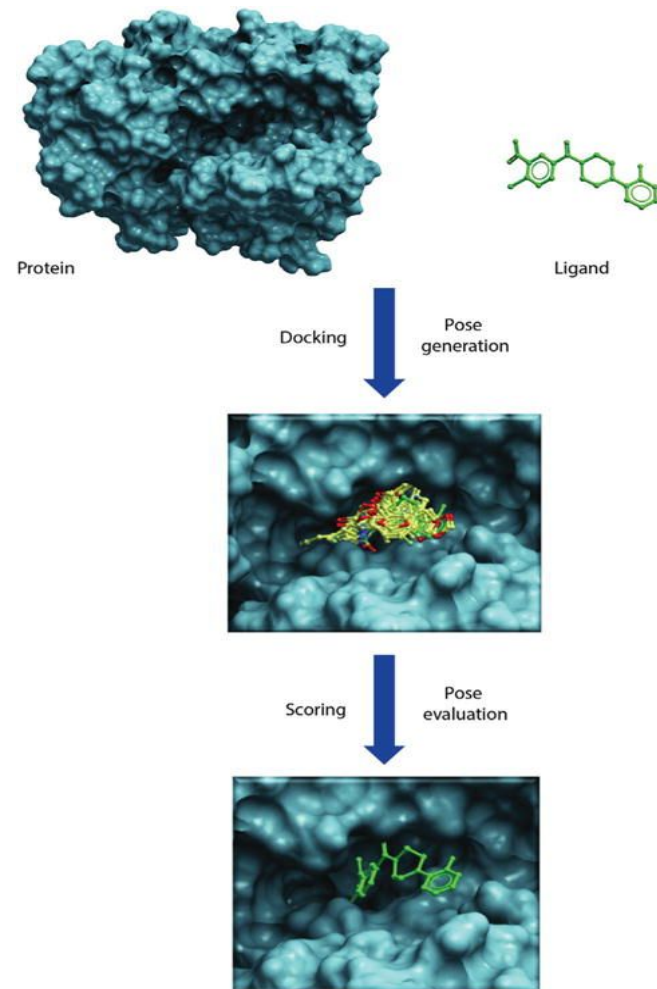
It is a computational method that predicts orientation of **ligand** when binds in an **active site** of target **receptor** to form stable complexes virtually.

Use-Reliable prediction of docking result is useful in structure-based drug designing.

Goal-To achieve optimized confirmation and relative orientation between ligand and receptor so that free energy of overall system is minimized.

It consist of two steps:

1. **Docking**- generation of receptor-ligand poses within binding cavity.
2. **Scoring**- Energetic evaluation of poses to find best scoring pose on the basis of minimum binding affinity or stability.



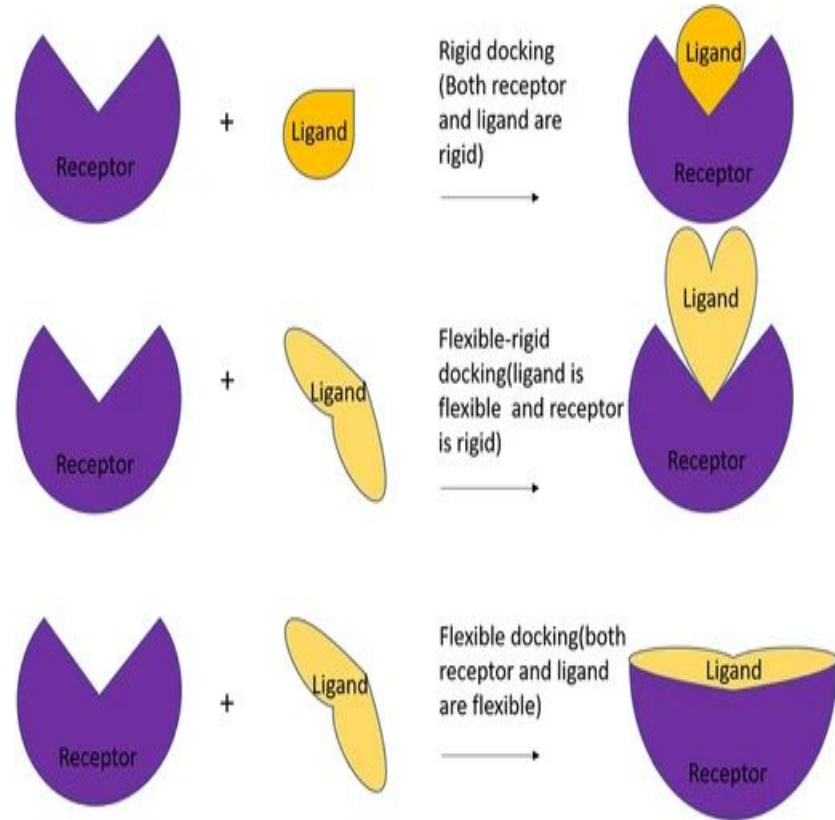
Components of Docking

- **Target Receptor**-macromolecules like protein, DNA, or RNA
- **Ligand**-small molecule(atom or ion) or drug candidates that bind to receptor.
- **Active Site**- it is specific region on the receptor where ligand binds and results in conformation change of receptor.

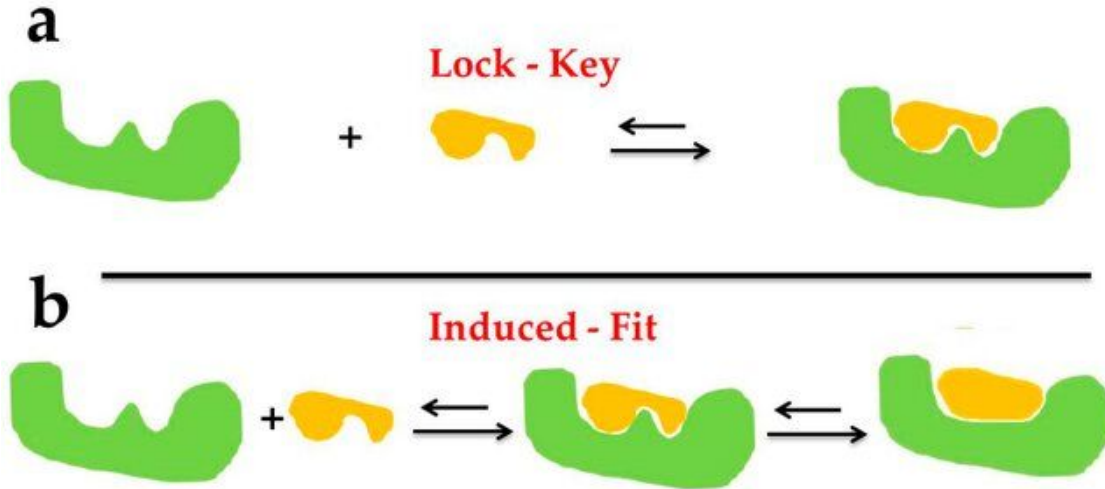
Classification of Docking

On the basis of :

- flexibility of the interacting molecules.
- Type of Interacting molecules.
- Active site

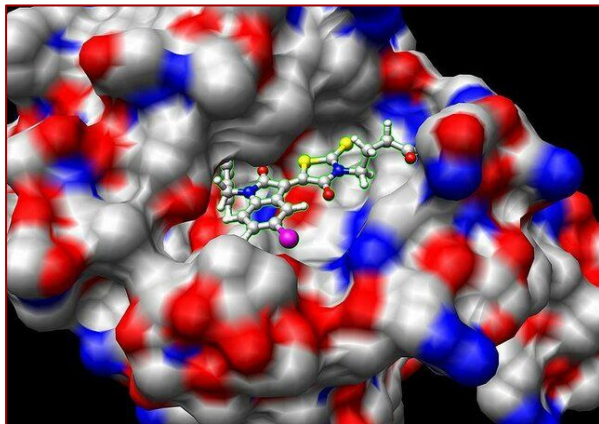


Types of Docking-on basis of flexibility



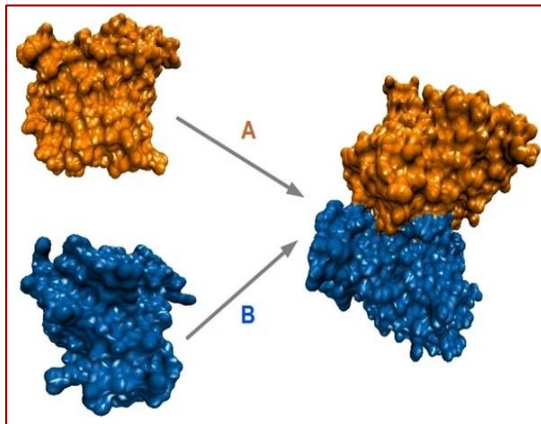
- **Rigid Docking(Lock and Key)**-The shape of both ligand and receptor is rigid and ligand acts as key to rotate and move to see if it fits in pocket of receptor as it is or not.Best for initial screening of thousands of molecules.
- **Flexible Docking(Induced Fit)**-it follows induced fit model where two molecules actually change shape dynamically(twist and bend).it has higher accuracy than rigid but much slower.

Types of Docking-Type of Interacting Molecules



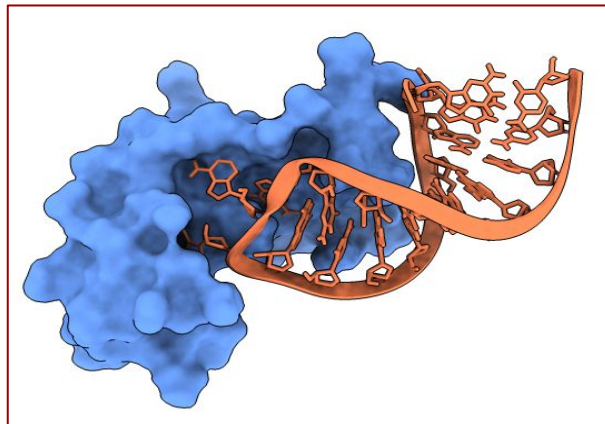
Protein-Ligand Docking-To predict how small molecule (drugs) bind to protein.

AutoDock Vina, AutoDock 4, GLIDE



Protein-Protein Docking- Modeling the interactions between two large proteins.

HADDOCK, ZDOCK



Protein-Nucleic Acid Docking-Modeling the interactions between proteins and DNA or RNA.

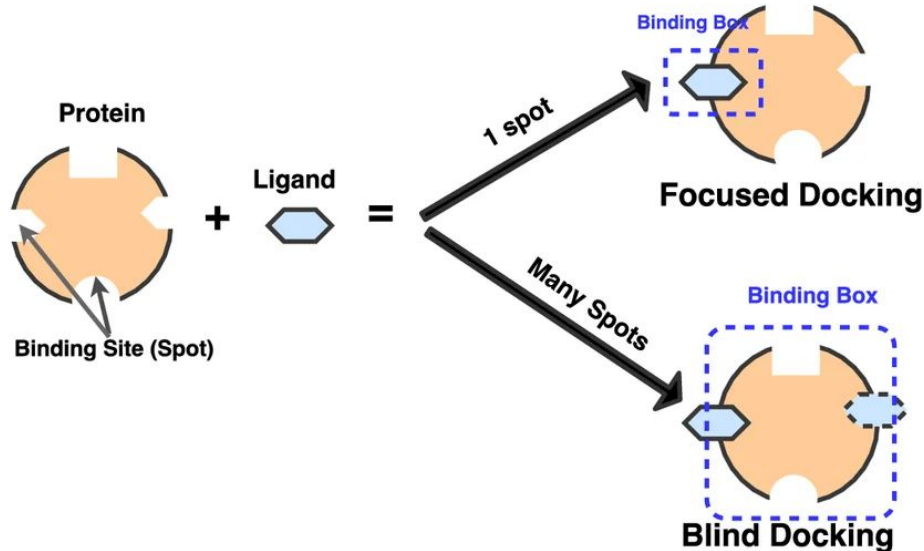
rDock, RLDOCK, XDock

Types of Docking-Active Site

Blind Docking-when active site on receptor is unknown so entire protein surface is scanned for potential binding pockets.

Active-site Docking- when active site is already known.

Inverse Docking- Dock single ligand against many different proteins to identify off-target effects.



Difference between Single Ligand Docking and Virtual Screening

Single Ligand Docking	Virtual Screening
Predicts optimal binding pose and affinity of a ligand within target receptor active site	Identify high-affinity ligand for target by searching high chemical space.
Already have known reference ligand	Ligand is unknown
Use case is Validation	Use case is Drug Discovery
Output is ranked ligand list	Output is detailed Interaction

Molecular Docking -Workflow



Requirements and Installations

1. MGL Tools(<https://ccsb.scripps.edu/mgltools/downloads/>)

Install version 1.5.7 for your OS(Windows, mac or linux)

2. OpenBabel (<https://github.com/openbabel/openbabel/releases/download/openbabel-3-1-1/OpenBabel-3.1.1-x64.exe>) for windows and
3. AutoDock Vina
4. Chimera
5. Protein receptor from [RCSB PDB](#) database
6. Ligands from [PubChem](#) database

File Formats

PDB:Protein Data Bank format file

SDF:Structure Data file

PDBQT:Protein Data Bank, Partial charge(Q), & Atom type(T) format

GPF:Grid Parameter File

GLG:Grid Log File

DPF:Docking Parameter File

DLG:Docking Log File

Step1: Retrieval of Target protein and Ligand

1. Download target protein receptor from PDB database
2. <https://www.rcsb.org/>
PDBID:3DTC
3. Download known ligand (C₂₇ H₂₄ N₂ O₃)

from PDB in 3D SDF format or from PubChem.

3DTC-Crystal structure of mixed-lineage kinase MLK1 complexed with compound 16 consist of chain A

RCSB PDB Deposit Search Visualize Analyze Download Learn About Careers COVID-19

251,217 Structures from the PDB archive 1,068,577 Computed Structure Models (CSM)

Enter search term(s), Ligand ID or sequence Include CSM

Advanced Search | Chemical Search | Browse Annotations

Structure Summary Structure Annotations Experiment Sequence Genome Ligands Versions

Biological Assembly 1

3DTC | pdb_00003dte

Crystal structure of mixed-lineage kinase MLK1 complexed with compound

PDB DOI: <https://doi.org/10.2210/pdb3DTC/pdb>

Classification: TRANSFERASE
Organism(s): Homo sapiens
Expression System: Escherichia coli
Mutation(s): Yes

Deposited: 2008-07-14 Released: 2009-03-31
Deposition Author(s): Fedorov, A.A., Fedorov, E.V., Meyer, S.L., Hudkins, R.L., Almo, S.C.

Experimental Data Snapshot
Method: X-RAY DIFFRACTION
Resolution: 2.60 Å
R-Value Free: 0.282 (Depositor), 0.270 (DCC)
R-Value Work: 0.218 (Depositor), 0.210 (DCC)
R-Value Observed: 0.221 (Depositor)

wwPDB Validation

Display Files Download Files Data API

FASTA Sequence
PDB/mmCIF Format
PDB/mmCIF Format (gz)
BinaryCIF Format (gz)
Legacy PDB Format
Legacy PDB Format (gz)
PDBML/XML Format (gz)
Structure Factors (CIF)
Structure Factors (CIF - gz)
Full Report

Metric
Rfactor
Clashscore
Ramachandran outliers
Sidechain outliers
RSRZ outliers

Validation Full (PDF - gz)
Validation (XML - gz)
Validation (CIF - gz)
Validation 2fo-fc coefficients (CIF - gz)
Validation fo-fc coefficients (CIF - gz)

Value
0.272
20
3.8%

Biological Assembly 1 (CIF - gz)
Biological Assembly 1 (PDB - gz)

3.7%

Explore in 3D Structure | Sequence Annotations | Electron Density | Validation Report | Ligand Interaction (VIN)

Global Symmetry: Asymmetric - C1
Global Stoichiometry: Monomer - A1

RCSB PDB PROTEIN DATA BANK 251,217 Structures from the PDB archive 1,068,577 Computed Structure Models (CSM)

Enter search term(s), Ligand ID or sequence Include CSM

Advanced Search | Chemical Search | Browse Annotations

PDB-101 PDB EMDatRelease NAKB PDB-REDO PDB-IHM

Display Files Download Files Data API

Definition (CIF Format)
Structure Data File (Ideal SDF)

12-(2-hydroxyethyl)-2-(1-methylethoxy)-13,14-dihydropyrido[3,4-c]carbazol-5(12H)-one

Created: 2008-07-24
Last modified: 2011-06-04

Find Related PDB Entry

1 entries where VIN is found as a standalone ligand

Find related ligands: Help

Similar Ligands (Stereospecific)

To get multiple ligands for target receptor,

download from PubChem in 3D SDF format.

Step2: Ligand and Receptor formats conversion via OpenBabel

Convert both ligand in SDF and protein in PDB into PDBQT formats.

Open cmd and give command :

```
obabel Ligand.sdf -O Ligand.pdbqt
```

```
obabel Receptor.pdb -O Receptor.pdbqt
```

Alternatively, Obabel GUI can also be used.

```
C:\Users\Diksha Marwaha\Downloads\Docking_tutorial>obabel Ligand.sdf -O Ligand.pdbqt
```

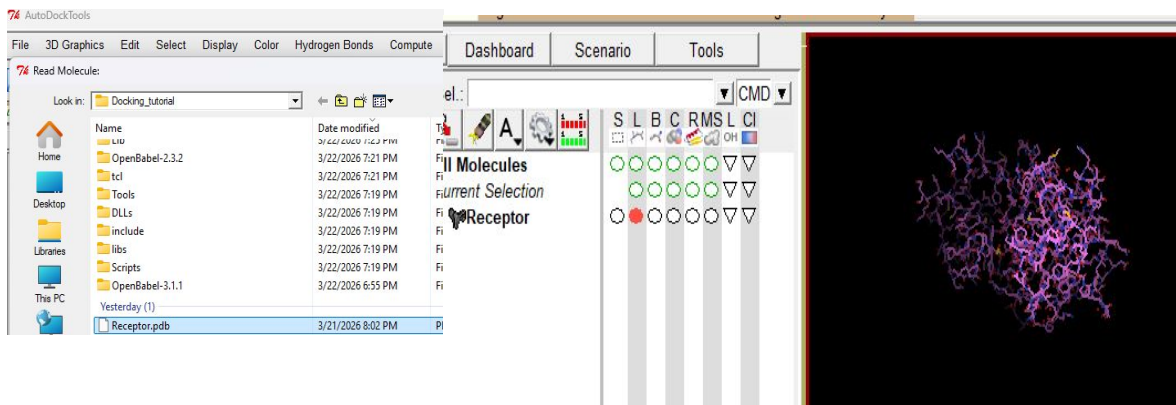
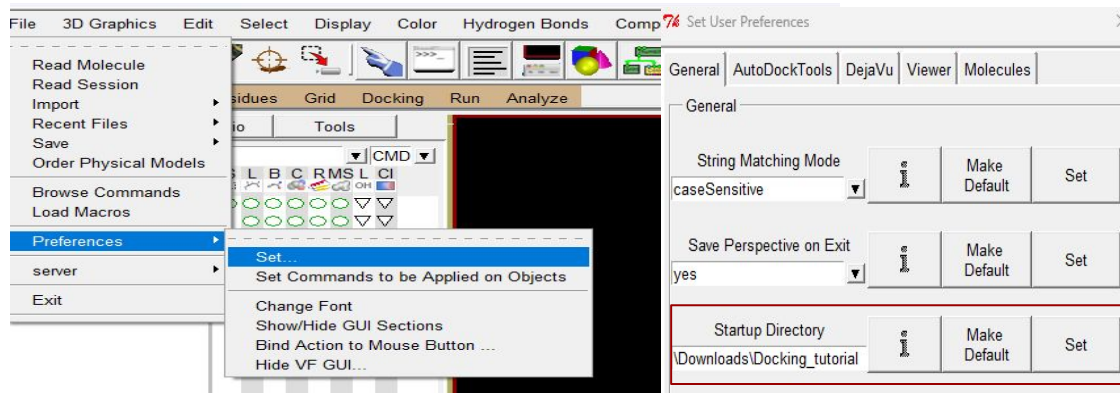
```
C:\Users\Diksha Marwaha\Downloads\Docking_tutorial>obabel Receptor.pdb -O Receptor.pdbqt
```

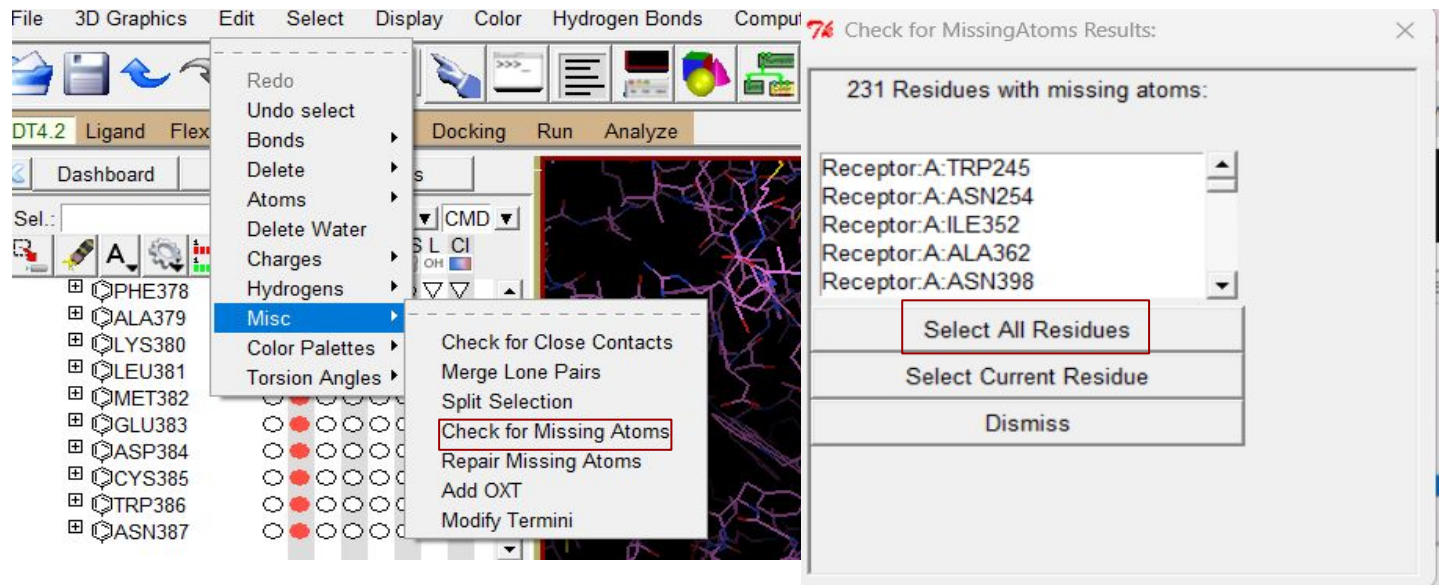
Step3: Target protein Preparation

1. Open AutoDock tools
2. Set the preferred working directory
3. File>read molecule>Receptor.pdb
4. **To remove water molecules and heteroatoms (inhibitors or ligands):**

Select chain A>Edit>Delete water

Select chain A hetero atom>Edit>delete selected atoms

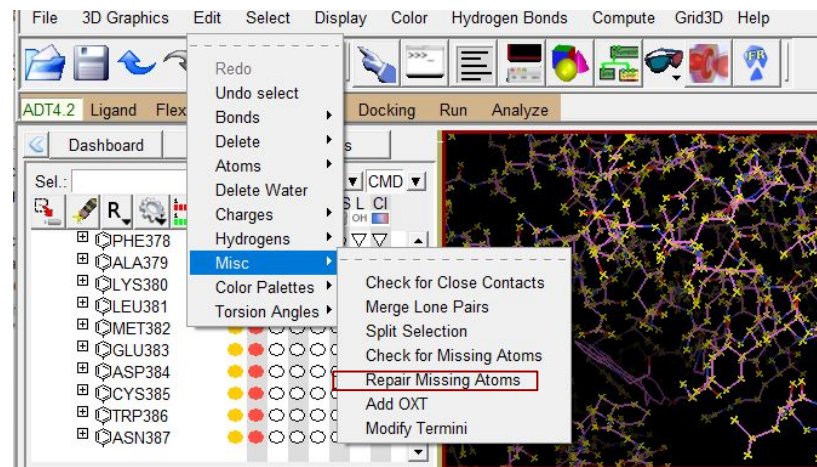


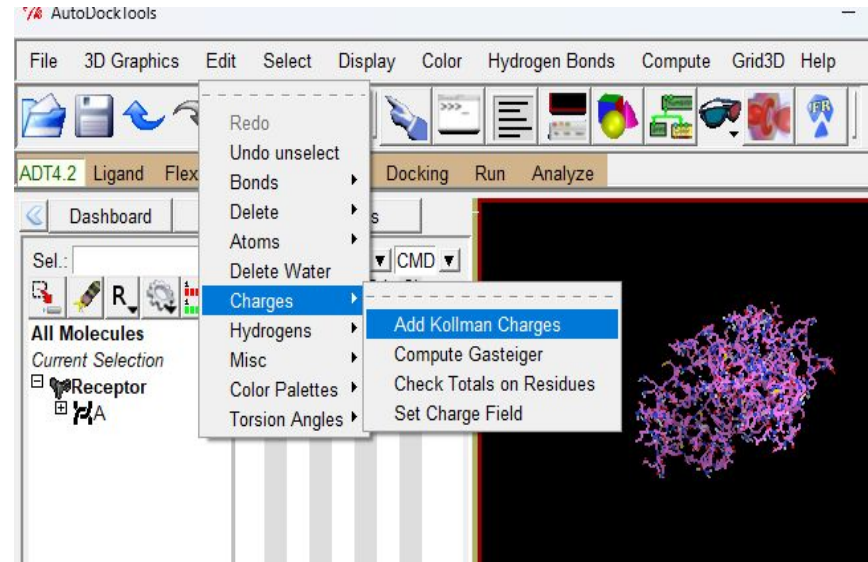
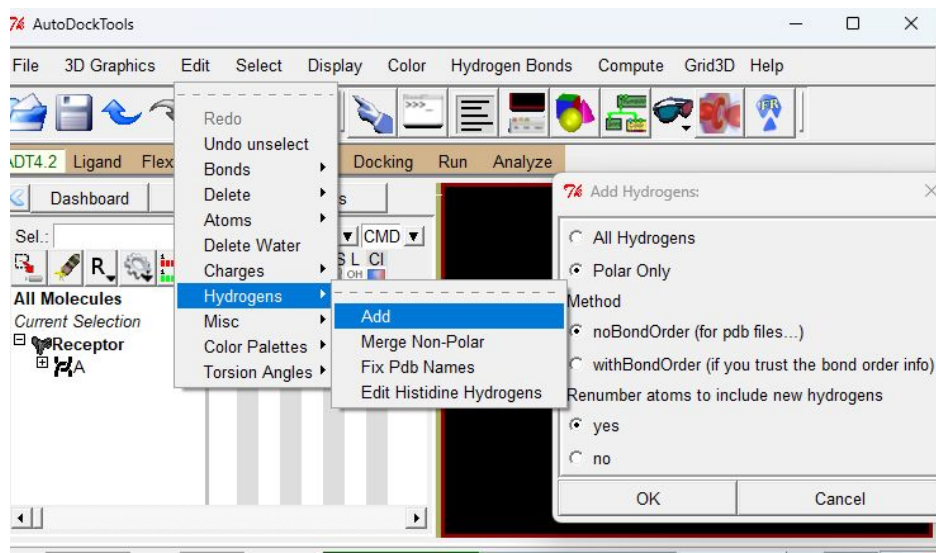


5. To prepare missing atoms:

Edit>misc>check for missing atoms

Edit>misc>Repair missing atoms>Save as 2 sets



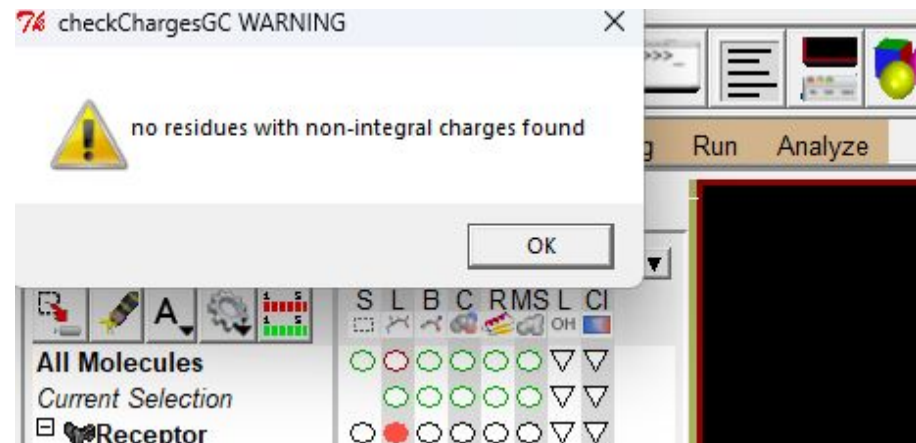
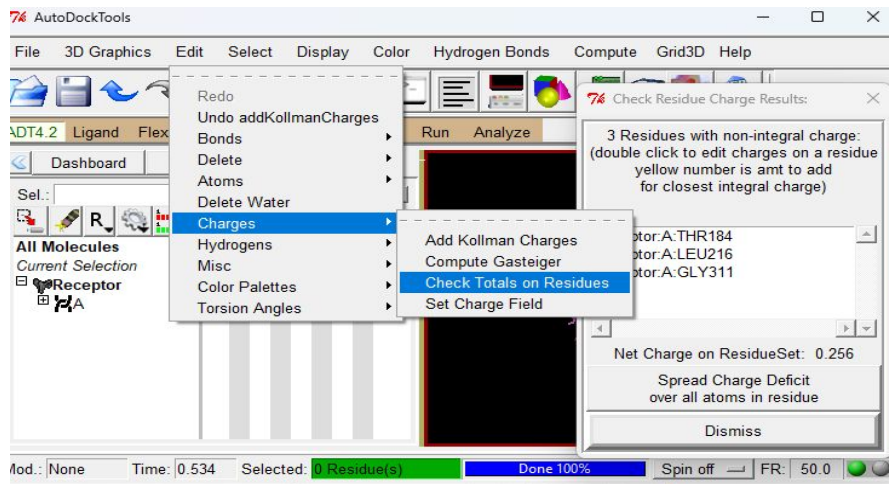


6.Add Hydrogen atoms:

Edit>Hydrogen>polar only

7. Add Charge:

Edit>charges>Add kollman charges



8. Add Charge:

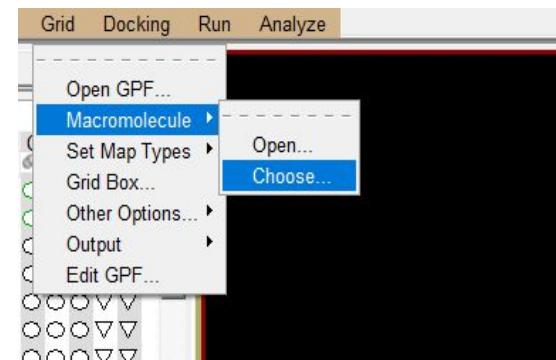
Edit>charges>check total on residues

9. Save receptor as PDBqt

Grid >

Macromolecule>choose>Receptor.pdbqt

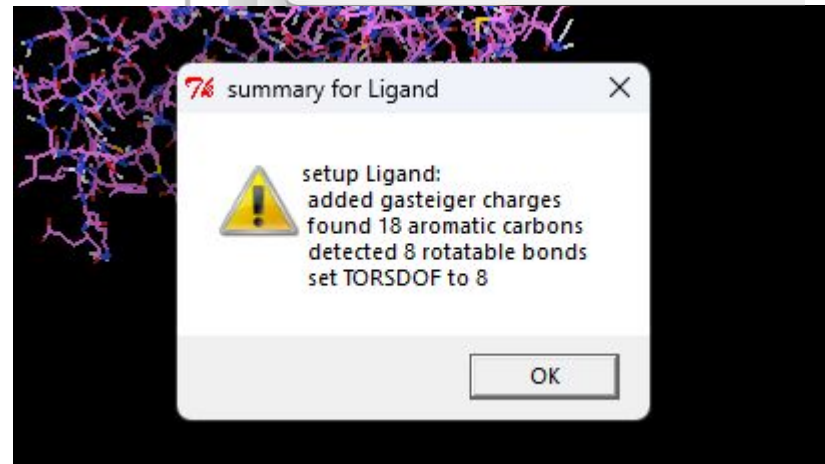
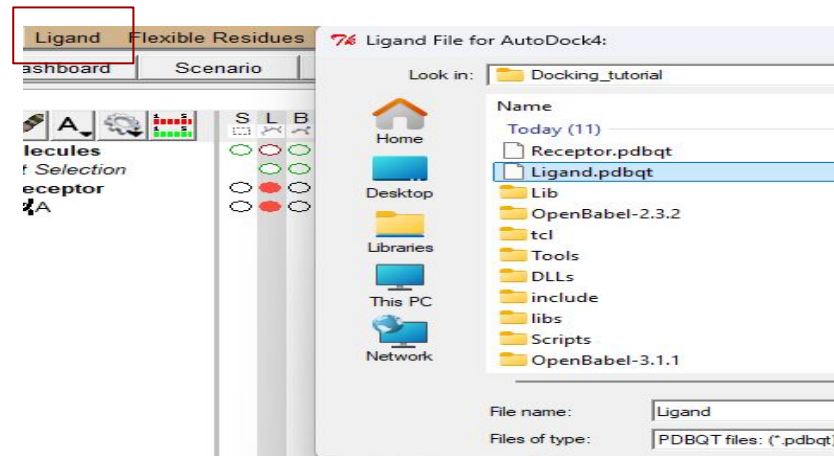
This means now the protein is completed charged.



Step4: Ligand Preparation

1.Ligand>Ligand.pdbqt

2.the ligand is already setup with charges.

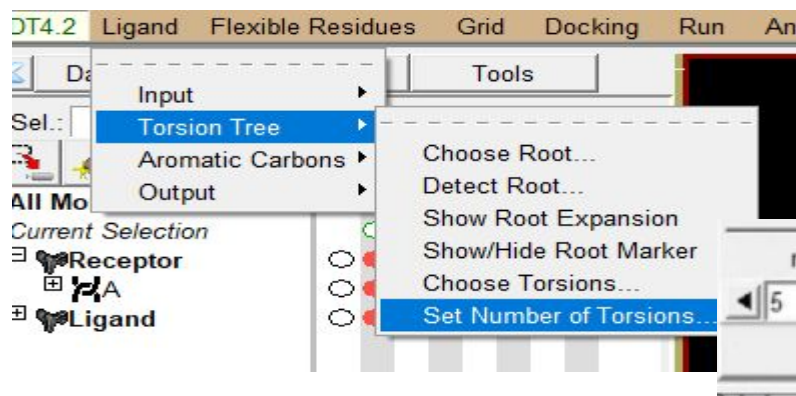
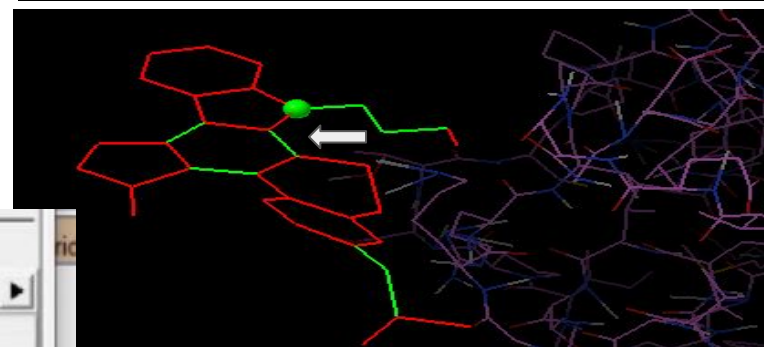
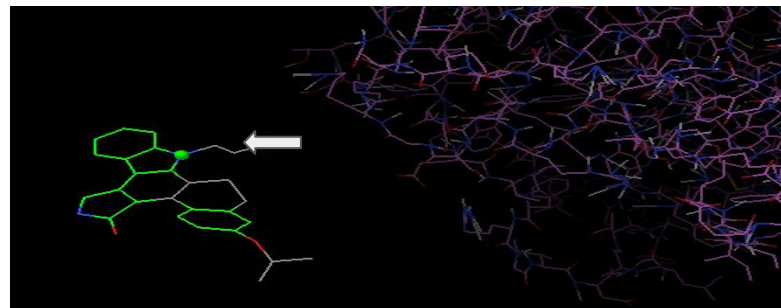
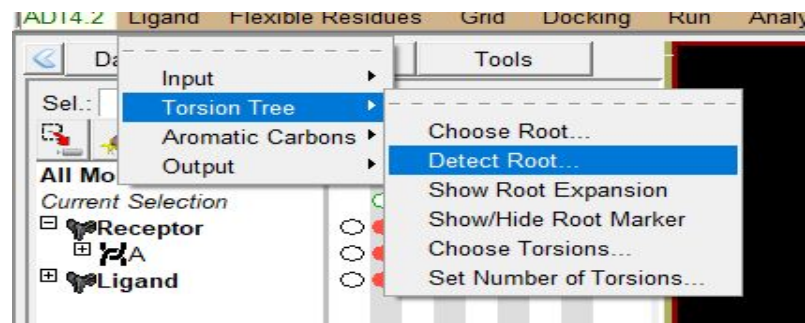


3.Ligand>Torsion tree> Detect root.

4.Ligand>Torsion tree>set no of Torsions

There are 5 no of active torsions in the ligand as shown by white arrow in figure.

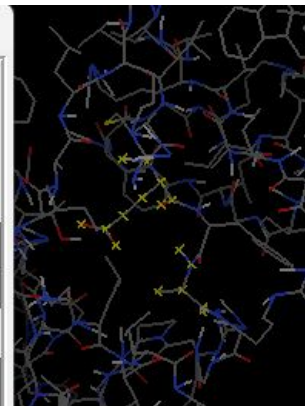
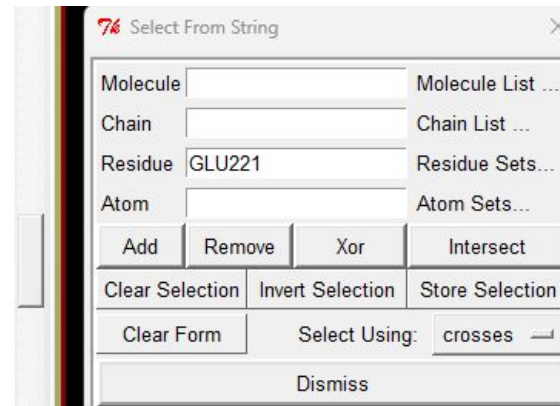
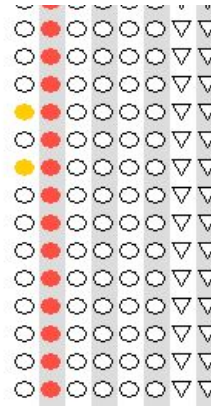
No of torsions means that no of rotatable bonds and that no of conformations will be generated or take more time for docking.



Step5: Active site

1. Get the Active site residues information from reported literature.
2. Reported residues were **GLU221**, **ALA223**, **ARG230**, **SER272** of receptor MLK1 for the ligand.
3. Select>Select from String>Residue>Add>Dismiss
4. Find X,Y,Z coordinates of active site of crystal structure(receptor.pdb)

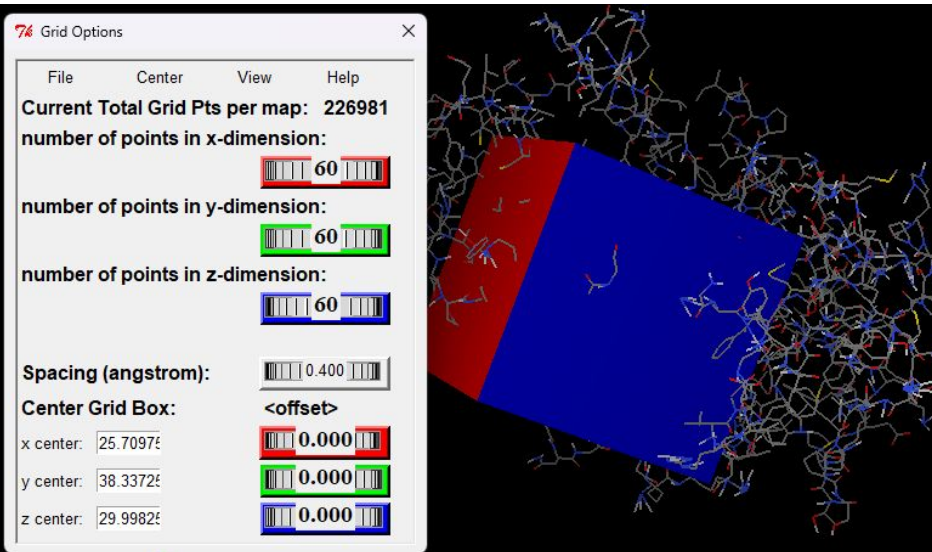
LEU218
VAL219
MET220
GLU221
PHE222
ALA223
ARG224
GLY225
GLY226
PRO227
LEU228
ASN229
ARG230
VAL231



Step5: Grid Generation

					X	Y	Z			
ATOM	579	OE2	GLU	A	221	25.945	45.067	22.98	1	32.42
ATOM	595	CB	ALA	A	223	25.367	38.703	26.074	1	30.05
ATOM	648	NH2	ARG	A	230	22.937	31.658	32.491	1	48.67
ATOM	981	OG	SER	A	272	28.59	38.55	38.448	1	68.4
						102.839	153.349	119.993		
						25.70975	38.33725	29.99825		

1. Copy and paste the last atom coordinates of active site amino acids, sum up each coordinate and divide it by number of residues
2. Grid>Macromolecule>choose Receptor.pdbqt>ok
3. Grid>Grid box.
4. Enter these X, Y, Z coordinates in the “centre Grid Box”
5. It will then exactly find location of Active site and covered by grid box,
6. File>Save as current.
7. Create config.txt file and save in same directory.



Step6: Docking via vina

1. Open cmd in windows and run :

```
vina --config config.txt --out  
output.pdbqt
```

2. Best affinity = -5.082 kcal/mol

```
receptor = Receptor.pdbqt  
ligand = Ligand1.pdbqt
```

```
center_x = 60  
center_y = 60  
center_z = 60
```

```
size_x = 25.70975  
size_y = 38.33725  
size_z = 29.99825
```

```
exhaustiveness = 8  
num_modes = 10
```

config.txt

mode	affinity	dist from best mode	
	(kcal/mol)	rmsd l.b.	rmsd u.b.
1	-5.082	0	0
2	-4.386	6.04	9.693
3	-4.38	6.639	10.41
4	-4.232	5.774	9.617
5	-4.222	6.327	10.7
6	-4.215	5.124	9.972
7	-4.11	5.98	9.818
8	-3.89	5.535	10.52
9	-3.851	5.607	8.712
10	-3.835	6.256	10.29

Step7: Visualization of docked complex via Chimera

Note :- Model with lowest binding affinity represents the most stable state. Because it estimates strength of binding between ligand and target protein represented as gibbs free energy in Kcal/mol. More negative the binding affinity, signifies more favourable binding.

In order to visualize all the conformations-

Open Chimera > Open receptor PDB > tools > structure binding analysis > ViewDock > open Output.pdbqt

